CAROM Frozen-GPT-2 Compiled-Channel Experiment

Detailed protocol, checkpoint results, causal interventions, and next-run design

CAROM research record

21 July 2026

Bottom line. A single frozen-GPT-2-small CAROM arm learned a moderately accurate in-distribution executor (0.412 slot accuracy at 4,000 updates) with strongly ordered-looking trajectories ($\tau = 0.904$). The natural trajectory became causally useful: replacing it by a shuffled schedule reduced accuracy from 0.412 to 0.297. This is meaningful mechanistic evidence, but accuracy is well below the matched TinyLM arm (0.709), and held-out length-5 transfer remains weak (0.214; 0.234 with a longer inference budget). A longer frozen-GPT-2 run is therefore justified as an optimization test, not yet as evidence of structural generalization or learned commutation.

1 Question and scope

The experiment asked whether natural-language command spans encoded by a *frozen* GPT-2-small model can drive a CAROM compiled heteroclinic channel. The task is synthetic compositional sequence transformation. Each example has six workspace slots containing symbols in $\mathbb{Z}_8$ and a program of two to five primitive transforms. Command descriptions are presented in scrambled surface order; the model must infer dependencies, produce a command-activity trajectory, mix reusable neural operators, and emit the final six symbols.

Training used program lengths 2–4. Length 5 was held out structurally. Thus:

- L2–4 accuracy measures in-distribution compositional execution;
- L5 accuracy measures extrapolation to a longer program;
- edge accuracy measures dependency-graph prediction;
- itinerary diagnostics describe the ordering of dominant command modes;
- trajectory replacements test whether the learned activity trajectory is causally relevant rather than merely correlated with the answer.

This was one seed and one frozen evaluation corpus. It is a mechanism screen, not a population estimate.

2 Architecture and training protocol

2.1 Language-conditioned compiler

GPT-2-small (124M parameters, hidden width 768, final hidden layer) was frozen. Span representations were produced using offset-mapped token spans. CAROM then used learned heads to (i) predict a directed dependency graph between command spans, (ii) compile that graph into generalized Lotka–Volterra inhibition, and (iii) route command representations onto a shared bank of neural workspace operators. Only the CAROM-side parameters were trained.

The workspace has six slots, vocabulary size eight, and up to five commands. The recurrent controller executes 72 internal steps at training/default inference. The endpoint objective is slotwise cross-entropy. The training loss also includes directed-edge binary cross-entropy and an entry-command term.

2.2 Exact run configuration

Seed	0
Training updates	4,000
Batch size	64
Optimizer	AdamW, learning rate 2×10^{-3}, weight decay 10^{-4}
Schedule	OneCycleLR over 4,000 updates
Gradient clipping	Global norm 1.0
Edge-loss coefficient	1.0; entry term coefficient 0.2
Checkpoints	0, 500, 1,000, …, 4,000
Frozen L2–4 corpus	256 examples; corpus seed 1234
Frozen L5 corpus	128 examples; corpus seed 5678
Hardware	RunPod Secure Cloud, one NVIDIA A100 SXM4 80GB
Image	`runpod/pytorch:1.0.2-cu1281-torch280-ubuntu2404`
Price and lifetime	USD 1.49/hr; approximately 2h09m including orchestration and repair

Training itself reached step 4,000 in 1,897 seconds (31.6 minutes). The original post-training pass then failed on malformed Python corpus unpacking. The repair loaded the missing `SP`, `E`, and `ENT` fields explicitly and added an `--interventions-only` continuation. It reused all retained checkpoints, evaluated nine checkpoints in about 40 seconds, and exited zero.

3 Evaluation instruments

Repaired itinerary panel. At each internal step, the dominant live command mode is recorded; consecutive duplicates are collapsed into phases. Coverage is the fraction of true command modes visited. Exact order requires the deduplicated visited sequence to equal the true topological order. Kendall-style τ scores concordant versus discordant rank pairs among visited modes. Because incomplete trajectories can still have high τ, tau must be read beside coverage and exact order.

Trajectory interventions. The natural learned trajectory is compared with three replacements on exactly the same frozen L2–4 examples:

- **forced**: true command order, equal dwell segments, amplitude 1.5;
- **shuffled**: an explicitly non-true command order with equal dwell;
- **smeared**: all live commands active simultaneously with equal mass.

Natural-minus-shuffled accuracy is a causal sensitivity diagnostic. The forced condition is not a perfect oracle: it changes amplitude and dwell statistics as well as order, so forced below natural is interpretable rather than contradictory.

Integration-budget sweep. The same L5 examples are evaluated with 72, 100, and 120 internal steps. A large increase would implicate insufficient settling time; a small increase leaves a structural or learned-computation limitation as the main explanation.

4 Checkpoint results

Table 1: Frozen GPT-2 checkpoint panel. Accuracy is mean slot accuracy.

Step	L2–4 acc	Edge acc	τ	Coverage	Exact	L5 acc	Nat−shuf
0	.139	.297	.040	.383	.000	.135	.000

Step	L2–4 acc	Edge acc	τ	Coverage	Exact	L5 acc	Nat−shuf
500	.206	.703	.477	.500	.043	.163	-.003
1000	.240	.703	.945	.695	.309	.176	.003
1500	.266	.703	.905	.682	.281	.145	-.007
2000	.291	.688	.862	.663	.246	.217	.025
2500	.331	.702	.871	.662	.262	.232	.060
3000	.372	.705	.892	.678	.273	.221	.079
3500	.402	.709	.904	.700	.285	.225	.089
4000	.412	.715	.904	.710	.297	.214	.115

Observed. Accuracy rose from 0.139 to 0.412. Edge accuracy jumped to about 0.70 by step 500 and then largely plateaued; endpoint accuracy continued improving. Itinerary tau rose sharply and remained near 0.86–0.90 after step 1,500, while coverage slowly recovered to 0.710. Exact full order reached only 0.297.

The high tau is therefore not equivalent to complete itinerary execution: the model orders the modes it visits fairly well but omits or fails to dominate with some modes. This distinction explains how tau can look strong while endpoint accuracy remains moderate.

5 Causal trajectory results

Table 2: Endpoint accuracy under natural and replaced trajectories.

Step	Natural	Forced	Shuffled	Smeared	Natural−shuffled
0	.139	.139	.139	.139	.000
500	.206	.209	.209	.208	-.003
1000	.240	.236	.238	.236	.003
1500	.266	.264	.273	.263	-.007
2000	.291	.272	.266	.285	.025
2500	.331	.278	.272	.264	.060
3000	.372	.311	.292	.308	.079
3500	.402	.317	.314	.318	.089
4000	.412	.309	.297	.301	.115

Observed. The intervention gap is essentially zero early and grows after step 2,000. At step 4,000, shuffling lowers accuracy by 11.5 percentage points relative to the natural trajectory. Smeared and forced schedules are similarly below natural. The learned trajectory has therefore become load-bearing for endpoint performance on the measured corpus.

Inferred. This rules out the strongest “decorative trajectory” account at the final checkpoint. It does not prove that every visited phase or every ordering detail is necessary, nor that the mechanism extrapolates beyond this task distribution.

6 Held-out length and integration depth

Table 3: L5 endpoint accuracy versus inference steps.

Step	Default L5	$S = 72$	$S = 100$	$S = 120$
0	.135	.135	.135	.135
500	.163	.163	.164	.163
1000	.176	.176	.180	.182
1500	.145	.145	.155	.168
2000	.217	.217	.233	.243
2500	.232	.232	.237	.249
3000	.221	.221	.224	.230
3500	.225	.225	.227	.224
4000	.214	.214	.224	.234

Observed. At the final checkpoint, adding 48 internal steps improves L5 accuracy by 2.1 points. The best measured L5 value (0.249) occurs at step 2,500 with $S = 120$, still far below in-distribution performance.

Inferred. Insufficient integration depth contributes slightly, but it is not the main L5 failure. The dominant limitation is learned computation/representation or structural extrapolation, not simply a short settling horizon.

7 Comparison with the retained TinyLM arm

The TinyLM comparison used the same nine checkpoint indices and intervention definitions, but its language encoder, batch/optimization details, and training artifact are not identical. It is contextual, not a controlled claim that one language model is intrinsically superior.

Final metric	TinyLM	Frozen GPT-2
L2–4 accuracy	.709	.412
Itinerary τ	.825	.904
Coverage	.911	.710
Natural−shuffled	.227	.115
L5 accuracy	.356	.214
L5 at $S = 120$	.362	.234

GPT-2 shows higher ordering correlation among visited phases but markedly lower coverage, accuracy, causal sensitivity, and length transfer. A plausible optimization explanation is that the CAROM routing heads must extract useful signals from a wider, frozen 768-dimensional space under the same 4,000-update budget. This is a hypothesis, not established by the current arm; a longer run is the direct test.

Importantly, these numbers do *not* establish learned commutation. Learned commutation predicts declining order sensitivity, which requires the repaired nonlinear swap-discrepancy and JVP commutator-energy probes. Here the observed natural-minus-shuffled gap grows, so the measured trajectory becomes more, not less, causally consequential over training.

8 What the experiment establishes

- **Observed.** A frozen GPT-2 span encoder can support learning above the random endpoint baseline in this CAROM compiled-channel architecture.
- **Observed.** Dependency prediction is learned early, but endpoint execution continues improving after edge accuracy plateaus.
- **Observed.** The final natural trajectory is causally useful relative to shuffled, smeared, and fixed-dwell forced replacements.
- **Observed.** High tau coexists with incomplete coverage and low exact-order rate; tau alone would overstate mechanism quality.

- **Observed.** Extra inference depth only modestly improves held-out L5 accuracy.
- **Not established:** robust GPT-2 semantic compilation, multi-seed reliability, learned commutation, schedule-uniform stability, or structural length generalization.

9 Higher-accuracy follow-up

The cleanest next test changes only the training budget: frozen GPT-2-small, seed 0, batch 64, losses and architecture unchanged, but 12,000 updates with a OneCycle schedule defined over all 12,000 updates and checkpoints every 1,000. This triples the number of training examples from 256,000 to 768,000 while preserving comparability. The same frozen corpora and intervention suite will be used. Primary success is final L2–4 accuracy above 0.55; a stronger target is 0.65. Mechanism gates remain coverage, exact order, natural-minus-shuffled, and L5. If accuracy rises while these degrade, the optimization succeeds only at the endpoint and the mechanism claim weakens.

Based on the measured 31.6-minute training time for 4,000 steps, expected training is about 95 minutes plus setup/checkpoint evaluation. The proposed guardrail is one Secure Cloud A100 SXM4 80GB at USD 1.49/hr, expected two hours (USD 2.98), hard cap three hours (USD 4.47), with auto-termination, immediate artifact retrieval and hash verification, then termination.

10 Provenance

Primary results are `projects/carom/experiments/20260721T222400Z-carom-gpt2-arm/`. The retrieved package contains nine checkpoints, nine per-checkpoint JSON files, `summary.json`, logs, repaired source, and a SHA-256 manifest covering 20 artifact files (105 MiB). The repaired runner SHA-256 is

```
211ae1387deec50c70d5f60b3571f17c
8e89d8ebc555763ef78d1be5d2ac8cc0
```

The compiled-channel source SHA-256 is

```
9171bcd065226d3f5ca96d7bb8f9f20d
9f144a11f251923382cbefb7be112223
```